

Vishal Bokare

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EDUCATION

Vishwakarma Institute Of Technology, Pune

2022 – 2026

B.Tech in Artificial Intelligence and Data Science — **CGPA: 8.56 (Final Year)**

Yeshwant Mahavidyalaya, Nanded

2019 – 2021

Higher Secondary Education in Science — **95.17%**

Mahatma Phule High School, Nanded

2018 – 2019

Secondary Education — **95.00%**

EXPERIENCE

UDNAV Software Services Private Limited

16 June 2024 – 15 December 2024

AI Software Development Intern - EduPlusCampus

- Engineered an AI-driven lesson planning platform using **Flask**, integrating **OCR**, **spaCy NLP**, **regex**, and **PDF processing** for automated academic document analysis.
- Developed backend APIs to parse academic calendars, extract syllabus topics, and generate semester-wise lesson plans based on institutional scheduling constraints.
- Implemented an intelligent scheduling algorithm that automatically assigned topics to lecture dates while considering holidays, exams, and academic events.
- Generated structured lesson plans in **Excel**, **PDF**, and **Word** formats, significantly reducing manual planning effort for educators.

PROJECTS

VanGuard | Machine Learning, Dijkstra, OpenCV, Arduino, ESP32

July 2024 – December 2024

- Engineered an AI-powered autonomous robotic navigation system using **ResNet18** and **PyTorch**, achieving **94% classification accuracy** across disaster events including fire, combat, destroyed buildings, military vehicles, and humanitarian aid zones.
- Developed a computer vision pipeline using **OpenCV** and **ArUco markers** for real-time localization, enabling precise robot positioning and navigation.
- Implemented **Dijkstra's shortest-path algorithm** for autonomous route planning, allowing the robot to efficiently deliver supplies to detected event locations.
- Integrated **ESP32**, IR sensors, and motor control modules with OpenCV-based image processing to execute real-time navigation based on AI-driven event classification.

FoundersCircle | Node.js, Express, MongoDB

January 2024 - May 2024

- Developed a scalable backend for a social networking platform using **Node.js**, **Express.js**, and **MongoDB**, enabling entrepreneurs to share startup ideas and build professional connections.
- Built secure **RESTful APIs** with **JWT authentication** and **bcrypt** password encryption, ensuring protected user access and session management.
- Designed optimized **MongoDB collections**, indexing strategies, and relationships for users, profiles, posts, comments, and follow systems to improve query performance.

AI-Powered Mock Interview System | RAG ,Groq LLM,Vector Embedding,Knowlegde Base,Answer Evaluation

Present

- Developed an AI-powered Mock Interview System that transforms unstructured resumes into a structured knowledge base and leverages **RAG with ChromaDB**, **Sentence Transformers**, and **Groq LLM** to generate personalized, context-aware technical interview questions.
- Engineered a stateful interview pipeline featuring adaptive question planning, **LLM-based answer evaluation**, and delivering structured feedback and dynamic interview progression based on candidate performance.

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, SQL, HTML/CSS

AI/ML: PyTorch, Scikit-learn, OpenCV, NLP, ResNet, CNN, Pandas, NumPy

Backend/Web: Node.js, Express, Flask, REST APIs, JWT Authentication

Databases: MongoDB, MySQL,Database Optimization, Indexing Strategies

DevOps/Tools: Git/GitHub, Linux, Postman, VS Code, IntelliJ IDEA

Relevant Coursework: Data Structures, OOP, DBMS, Operating Systems, Machine Learning, Deep Learning

PUBLICATIONS & CERTIFICATIONS

- VANGUARD: Autonomous Robotic Navigation in Conflict Zones** — Patent No. **202421074986 A**, Published on **15/11/2024**.
- Utilizing IoT: Farm Automation Using NodeMCU** — Presented at **ICABEC-2023**.
- Java** Certified by **Udemy 2023**.
- Machine Learning** Certified by **NPTEL-2024**.

EXTRACURRICULAR

- Volunteer** — Ganpati Festival: Served as a **Police Mitra**, assisting police staff in crowd management, safety support, and social welfare activities.